

DIEGO RAMOS AGUILERA, Ph.D.

Interdisciplinary researcher and engineer specializing in climate vulnerability and sustainable energy
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RESEARCH INTERESTS

My research explores how extreme climate hazards, energy systems, and the built environment interact across space and time to shape human livelihoods. My previous work has focused on understanding extreme heat and how it affects thermal comfort, residential energy use and environmental justice, drawing on perspectives and methods from engineering, geography and data science. I am particularly interested in interdisciplinary approaches to examine hazard vulnerability and climate adaptation, as well as in solutions and technologies seeking to promote equitable and sustainable development.

EDUCATION

Ph.D., Environmental Engineering University of Southern California	August 2021 — August 2026
M.S., Green Technologies University of Southern California. <i>3.9 GPA</i>	August 2021 — December 2024
M.Sc., Environment and Development King's College London. <i>Pass with Distinction</i>	September 2019 — September 2020
B.Eng., Mechanical and Energy Engineering Brunel University London. <i>1st Class Honors</i>	September 2015 — May 2019

TECHNICAL STRENGTHS

- Experimental design and collection of environmental data
- Analytical skills and statistical modeling
- Data engineering with Python (including libraries such as *scipy*, *statsmodels*, *geopandas*, *xarray*, and *dask*)
- Handling of large datasets and high-performance computing (experience in *bash*)
- Spatial analysis and geographic information systems (e.g., ArcGIS, QGIS)
- Data visualization and research communication

RESEARCH EXPERIENCE

Research Assistant <i>Department of Civil and Environmental Engineering, University of Southern California</i> Project: Characterizing extreme heat vulnerability in the context of environmental, social, and technological systems in urban settings Advisors: Dr. Kelly Sanders, Dr. Jiachen Zhang	August 2022 — June 2026
Research Assistant <i>Department of Civil and Environmental Engineering, University of Southern California</i> Project: Impacts of solar-reflective coating on pre-K-12 schoolyards at the Los Angeles Unified School District (LAUSD) Advisor: Dr. Jiachen Zhang	August 2025 — March 2026
Master's Research <i>School of Social Science and Public Policy, King's College London</i>	January 2020 — August 2020

Project: Quantifying vulnerability to environmental hazards in southern Madagascar: An urban case study in the city of Toliara
Advisor: Dr. Faith Taylor

Undergraduate Dissertation

January 2020 — August 2020

Institute of Energy Futures, Brunel University London

Project: Investigation of cool and green roofs and façades to reduce energy demand in a student accommodation building at Brunel University

Advisor: Dr. Maria Kolokotroni

TEACHING EXPERIENCE

Teaching Assistant, Introduction to Civil Engineering Graphics Spring 2026

Department of Civil and Environmental Engineering, University of Southern California

Teaching Assistant, Energy Systems and Environmental Tradeoffs Fall 2023, 2025

Department of Civil and Environmental Engineering, University of Southern California

Assistant Lecturer, Introduction to Energy Fall 2024

Department of Civil and Environmental Engineering, University of Southern California

Teaching Assistant, Statics and Dynamics Fall 2022

Department of Civil and Environmental Engineering, University of Southern California

PROFESSIONAL EXPERIENCE

WELL® Building Standard Intern June 2019

Aguilera Ingenieros, Madrid (Spain)

Powertrain Solutions Engineering Intern August 2017 — August 2018

Bosch, London (United Kingdom)

Assistant Community Manager January 2013 — July 2018

Fundación Agua de Coco, Madrid (Spain)

SERVICE AND VOLUNTEERING

2024 — present **Reviewer** for *Environmental Research: Health, Environmental Research Letters* and *Environmental Research: Climate* journals.

2023 **Volunteer** for the Climate Leadership Conference, hosted by The Climate Registry in Los Angeles (CA), May 10-12.

2022 — 2024 **Member and Student Advisor** for USC Student Sustainability Committee (Los Angeles, CA, USA).

2014 — 2019 **Summer Volunteer** for ONG Bel Avenir (Toliara, Madagascar).

PEER-REVIEWED JOURNAL PUBLICATIONS

1. Fazelvalipour, S., He, Y., Badami, M.M., **Ramos Aguilera, D.**, Ramirez, V., Aghaei, Y., Farahani, V.J., Huang, Y., Chen, M., Forestieri, S., Quiros, D., Sioutas, C., and Zhang, J. (202X). Forgone Community Emission Benefits in Southern California Without Zero-Emission Truck Regulations. *Environmental Pollution*, under review.
2. **Ramos Aguilera, D.**, Wang, Y., Zhang, L., Zhang, J., Sanders, K.T. (2026). Assessing disparities in air conditioning operation in Southern California. *Nature Communications*, under review. <https://doi.org/10.21203/rs.3.rs-10103387/v1> [preprint].

3. **Ramos Aguilera, D.**, Shreevastava, A., Olivier, A., Sanders, K.T., Zhang, J. (2026). Incorporating humid heat metrics and air conditioning ownership into heat vulnerability assessments. *City and Environment Interactions*, under review.
4. Zhang, X., Peplinski, M., **Ramos Aguilera, D.**, Sanders, K.T., Ebel, S., Scovronick, N., Kalmus, P., Chang, H.H. (2025). Smart meter-derived air conditioning penetration rate modifies the association between ambient temperature and emergency department visits: a ZIP code-level analysis in southern California. *Epidemiology*, 37(5), p. 719-723. <https://doi.org/10.1097/EDE.0000000000002006>.
5. Shreevastava, A., Hulley, G., Prasanth, S., Chakraborty, T., **Ramos Aguilera, D.**, Sanders, K.T., Yin, Y. (2025). Contemporary income inequality outweighs historic redlining in shaping intra-urban heat disparities in Los Angeles. *Nature Communications* 16, 4950. <https://doi.org/10.1038/s41467-025-59912-x>.

CONFERENCE POSTER AND ORAL PRESENTATIONS

* indicates presenting author other than Ramos Aguilera, D.

1. **Ramos Aguilera, D.**, Zhang, L., Zhang, J., Habre, R., Sanders K.T.* (2026, April 15). Quantifying the influence of social and environmental factors on residential air conditioning in Southern California [poster]. *NIH Health and Extreme Weather Program: Solutions and Community Resilience Science Symposium*, Rockville, MD, United States.
2. **Ramos Aguilera, D.**, Wang, Y., Zhang, J., Sanders, K.T. (2025, December 15). Assessing the disparities in air conditioning use for heat management across Southern California [poster]. *American Geophysical Union (AGU) Annual Meeting 2025*, New Orleans, LA, United States.
3. Shreevastava, A.*, Hulley, G., Prasanth, S., Chakraborty, T., Ramos Aguilera, D., Sanders, K.T., Yin, Y. (2025, December 15). The roots and remedies of urban heat inequity: A remotely sensed exploration of diurnal and seasonal thermal disparities in Los Angeles [poster]. *American Geophysical Union (AGU) Annual Meeting 2025*, New Orleans, LA, United States.
4. **Ramos Aguilera, D.**, Zhang, J., Sanders, K.T. (2025, April 11). Understanding adaptive capacity to extreme heat in Southern California [oral]. *Southern California Civil and Environmental Engineering Research Symposium (SoCal CEERS)*, Irvine, CA, United States.
5. Ramirez, V.*, **Ramos Aguilera, D.**, Fazlvalipour, S., Zhang, J. (2025, April 10). Quantifying racial and ethnic inequities for vehicle miles traveled (VMT) and truck emission disparities in disadvantaged communities: A literature review [poster]. *Southern California Civil and Environmental Engineering Research Symposium (SoCal CEERS)*, Los Angeles, CA, United States.
6. **Ramos Aguilera, D.**, Zhang, J., Sanders, K.T. (2024, December 9). Integrating sociodemographic, health, and environmental data to assess the spatiotemporal trends of vulnerability to heatwaves in California [oral]. *American Geophysical Union (AGU) Annual Meeting 2024*, Washington, D.C., United States.
7. **Ramos Aguilera, D.**, Olivier, A., Shreevastava, A., Sanders, K.T. (2023, December 12). Identifying hotspots of heat risk in Southern California through coupled exposure and vulnerability metrics [oral]. *American Geophysical Union (AGU) Fall Meeting 2023*, San Francisco, CA, United States.
8. Shreevastava, A.*, Hulley, G.C., Prasanth, S., Yin, Y., Chakraborty, T., **Ramos Aguilera, D.**, Peplinski, M., Sanders, K.T. (2023, December 11). Unequal heat exposure in urban areas: unraveling the role of historic redlining and present-day inequalities in Los Angeles [oral]. *American Geophysical Union (AGU) Fall Meeting 2023*, San Francisco, CA, United States.
9. **Ramos Aguilera, D.**, Jin, A.S., Sanders, K.T. (2023, November 13). Equitable access to cooling centers in the context of heat vulnerability in LA county [poster]. *Behavior, Energy, and Climate Change (BECC) Conference*, Sacramento, CA, United States.

10. **Ramos Aguilera, D.**, Olivier, A., Sanders, K.T. (2023, October 20). Assessing vulnerability and exposure to heat across Southern California [poster]. *Southern California Civil and Environmental Engineering Research Symposium (SoCal CEERS)*, Los Angeles, CA, United States.
11. **Ramos Aguilera, D.**, Sanders, K.T. (2022, December 13). A high-spatial resolution analysis of heat stress in the Los Angeles region using WRF [oral]. *American Geophysical Union (AGU) Fall Meeting 2022*, Chicago, IL, United States.
12. **Ramos Aguilera, D.**, Sanders, K.T. (2022, September 23). Unlocking the potential of WRF models for high-resolution heat stress analysis [poster]. *Astani Research Symposium*, Los Angeles, CA, United States.

OTHER PUBLICATIONS

1. **Ramos Aguilera, D.** (2021, June 11). What the pandemic taught us about remote research. *King's College London*. <https://www.kcl.ac.uk/what-the-pandemic-taught-us-about-remote-research>

FELLOWSHIPS AND GRANTS

2022, 2024	Das Family Travel Award
2021 — 2022	USC Viterbi School of Engineering Doctoral Fellowship
2015 — 2019	Brunel University London Academic Excellence Scholarship

AWARDS

2025	Winner of the 2025 Data Visualization Competition, <i>American Geophysical Union (AGU) GeoHealth Early Career Research Committee</i>
2025	First Place Oral Presentation, <i>Southern California Civil and Environmental Engineering Research Symposium (SoCal CEERS)</i>
2020	Department of Geography Best Thesis Award, <i>King's College London</i>
2019	Cork Crown and Seal Prize, <i>Brunel University London</i>

LICENSES AND CERTIFICATIONS

2025	Trusted Reviewer Status, <i>Institute of Physics (IOP)</i>
2023	Human Research - Social-Behavioral Human Subjects, <i>CITI Program</i> [expired: 2026]
2021	Responsible Conduct of Research, <i>CITI Program</i>
2021	Data Translator, <i>Workera</i>
2019	WELL Accredited Professional, <i>International WELL Building Institute</i> [expired: 2021]

AFFILIATIONS

2026 —	Member, <i>Esri Young Professionals Network (YPN)</i>
2022 — 2025	Student Member, <i>American Geophysical Union (AGU)</i>
2019 — 2020	Student Member, <i>Energy Institute</i>
2016 — 2019	Affiliate (Student) Member, <i>Institute of Mechanical Engineers (IMechE)</i>